

ENTHUSIAST, LADER & ACHIEVER COURSE

PHASE : (All Phase)

TARGET : PRE-MEDICAL 2020

Test Type : DRILL TEST # 06

Test Pattern : NEET (UG)

TEST DATE : 09-04-2020

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	3	3	2	2	2	2	3	2	1	4	3	2	4	2	1	3	3	1	2	4	3	3	2	1	2	2	1	2	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	2	2	2	1	4	3	3	4	4	3	2	1	1	3	4	1	1	2	3	4	3	3	4	4	4	3	2	2	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	2	2	3	3	3	3	1	4	3	3	4	3	2	3	2	1	3	3	1	4	2	1	4	1	1	1	3	2	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	2	3	2	4	3	3	1	4	2	3	1	2	4	4	3	2	1	1	4	4	2	1	4	1	2	4	4	2	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	1	4	3	4	1	3	2	3	3	2	3	4	4	1	2	1	1	4	1	3	4	1	4	1	2	1	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	1	4	3	3	2	1	2	4	1	1	4	2	1	4	4	2	2	4	4	2	2	4	2	1	3	3	2	3	2

HINT - SHEET

1. **Ans (4)**

$$E_{C \rightarrow A} = E_{C \rightarrow B} + E_{B \rightarrow A}$$

$$\frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$$

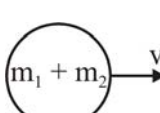
$$\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

2. **Ans (3)**

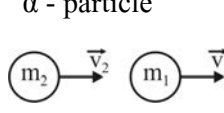
For Laminar flow streamline should be parallel and never intersect with each other.

3. **Ans (3)**

α - particle



Before emitting
 $\vec{p}_i = (m_1 + m_2) \vec{v}$
 in horizontal direction



after emitting
 $\vec{p}_f = m_1 \vec{v}_1 + m_2 \vec{v}_2$
 in horizontal direction

by conservation of L.M.

$$\vec{p}_i = \vec{p}_f$$

$$(m_1 + m_2) \vec{v} = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

Thus $\vec{v}$ must be parallel to $(m_1 \vec{v}_1 + m_2 \vec{v}_2)$

$$\text{As } \vec{v} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{(m_1 + m_2)}$$

5. **Ans (2)**

$$\int_{V_A}^{V_B} dv = - \int_4^6 \vec{E} \cdot d\vec{r}$$

$$V_B - V_A = - \int_{V_A}^{V_B} (20\hat{i}) \cdot (dx\hat{i}) = -40 \text{ Volt}$$

6. **Ans (2)**

$$v_{\text{rel}} = v_1 + v_2$$

$$a_{\text{rel}} = (a_1 + a_2)$$

for relative motion

$$v_{\text{rel}}^2 = u_{\text{rel}}^2 + 2a_{\text{rel}} \cdot S_{\text{rel}}$$

for max. initial separation $v_{\text{rel}} = 0$

$$0^2 = (v_1 + v_2)^2 + 2 \{-(a_1 + a_2)\} (\ell_{\text{max}} - 0) = 0$$

$$\ell_{\text{max}} = \frac{(v_1 + v_2)^2}{2(a_1 + a_2)}$$

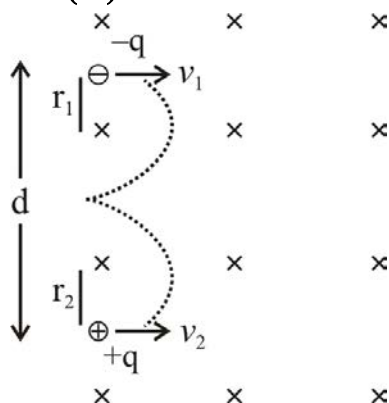
7. **Ans (2)**

F = slope of p - t curve

at t = 3,

slope = -1.

8. Ans (3)



for particle to not collide

$$(2r_1 + 2r_2) < d$$

$$2 \left(\frac{mv_1}{qB} + \frac{mv_2}{qB} \right) < d$$

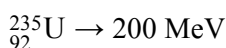
$$d = \frac{2m}{qB} (v_1 + v_2)$$

9. Ans (2)

$$\begin{aligned} y &= (A + B) \cdot (\overline{A} \cdot \overline{B}) \\ &= (A + B) \cdot (\overline{A} + \overline{B}) \\ &= A\overline{A} + A\overline{B} + B\overline{A} + B\overline{B} \\ &= 0 + A\overline{B} + B\overline{A} + 0 \\ &= \text{XOR Gate} \end{aligned}$$

10. Ans (1)

Energy in one reaction of



in N_A Reaction $E = 200 \text{ MeV} \times N_A$

means energy emitted by 235 gram



$$\text{by } m \text{ gram uranium} \rightarrow \frac{(N_A \times 200 \text{ MeV}) \times m}{235}$$

If it is in one day then

$$\frac{(N_A \times 200 \text{ MeV}) m}{235} = (0.95 \text{ Mw}) \times 1 \text{ day}$$

$$m = \frac{0.95 \text{ MW} \times (24 \times 3600 \text{ sec}) \times 235}{N_A \times 200 \text{ MeV}}$$

$$m = \frac{0.95 \times 10^6 \times 24 \times 3600 \times 235}{6 \times 10^{23} \times 200 \times 10^6 \times 1.6 \times 10^{-19}} \text{ gram}$$

on solving

$$m = 1 \text{ gram}$$

11. Ans (4)

$$\begin{aligned} h &= \frac{2\rho_T \cos \theta}{\rho_{rg}} h = \frac{2\rho_T \cos \theta}{\rho_{rg}} \\ h &\propto \frac{1}{g_{\text{eff}}} \end{aligned}$$

12. Ans (3)

W = area of cycle

$$W = \frac{\pi}{2} ab$$

$$\text{From diagram } 2a = 40 \Rightarrow a = 20$$

$$b = 20 \Rightarrow b = 20$$

$$W = \frac{\pi}{2} (20) (20) = 200 \pi \text{ J}$$

13. Ans (2)

There is no resistor in series with capacitor,
Hence time constant

$$\tau = RC = 0$$

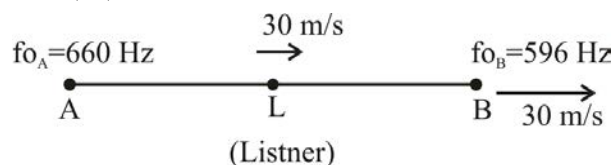
14. Ans (4)

$$\tan \theta = \frac{v^2}{Rg}$$

$$\tan 45^\circ = \frac{v^2}{(90)(10)}$$

$$v = 30 \text{ m/s}$$

15. Ans (2)



Apparent frequency for whistle by 'A'

$$(f_{\text{app}})_A = f_{0A} \left(\frac{V - V_L}{V - V_S} \right)$$

$$= 660 \left(\frac{330 - 30}{330 - 0} \right) = 600 \text{ Hz}$$

$$(f_{\text{app}})_B = f_{0B} \left(\frac{V - V_L}{V - V_S} \right)$$

$$= 596 \left[\frac{330 - (-30)}{330 - (-30)} \right]$$

$$= 596 \text{ Hz}$$

$$\text{Beat frequency } f_b = (f_{\text{app}})_A - (f_{\text{app}})_B$$

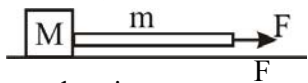
$$= 600 - 596$$

$$= 4 \text{ beats/sec}$$

16. **Ans (1)**

Since $PV = RT$ and $T = \text{constant}$; $\therefore PV = \text{constant}$.

17. **Ans (3)**



$$\text{acceleration } a = \frac{F}{m + M}$$

force applied on block = Ma

$$= \frac{MF}{m + M}$$

18. **Ans (3)**

$$(A) P = n \frac{hc}{\lambda}$$

$$10^{-6} \times (10^{-10}) = n \frac{12400}{5600} \times 1.6 \times 10^{-19}$$

$$\Rightarrow n = 300$$

19. **Ans (1)**

Due to D_3 , effective resistance of circuit is R

$$i = \frac{E}{R}$$

20. **Ans (2)**

$$[v] = [T^{-1}]$$

$$[\ell] = [L^1]$$

$$[F] = [M^1 L^1 T^{-2}]$$

$$m = \frac{\rho^2 F}{4v^2 \ell^2}$$

$$[m] = \frac{[F]}{[v^2][\ell^2]} = \frac{[M^1 L^1 T^{-2}]}{[T^{-2}][L^2]}$$

$$[m] = [M^1 L^{-1}]$$

21. **Ans (4)**

$$K.E_i = P.E_f + W.D_{fr}$$

$$\frac{1}{2}(1)(20)^2 = (1)(10)(18) + W.D_{fr}$$

$$200 = 180 + W.D_{fr}$$

$$W.D_{fr} = 20 \text{ J}$$

22. **Ans (3)**

$$I = MK^2$$

$$\frac{7}{5}MR^2 = MK^2$$

$$K = \sqrt{1.4} R$$

23. **Ans (3)**

$$V_1 = \frac{k(4q)}{r}, V_2 = \frac{k(-2q)}{r}$$

$$V_3 = \frac{k(2q)}{r}, V_4 = \frac{k(2q)}{r}$$

$$\Rightarrow V_1 > V_3 = V_4 > V_2$$

24. **Ans (2)**

from graph

$$F = -\frac{27}{3}x$$

$$(1) a = -9x$$

$$a = -9x$$

$$w^2 = 9 \Rightarrow w = 3$$

$$\frac{2\pi}{T} = 3$$

$$\Rightarrow T = \frac{2\pi}{3} \text{ sec}$$

25. **Ans (1)**

$$E' = \frac{E}{20} \left(\frac{\ell_0}{10} \right) (10) \quad \dots\dots(1)$$

When cross sectional area is doubled

$$E' = \frac{E}{15} \frac{\ell'}{10} (5) \quad \dots\dots(2)$$

From (1) & (2)

$$\frac{\ell_0}{20} = \frac{\ell'}{30}$$

$$\ell' = \frac{3\ell_0}{2} = 1.5\ell_0$$

26. **Ans (2)**

$$I_D = \frac{dq}{dt} = \frac{d}{dt} (VC) = C \frac{dV}{dt}$$

$$\text{Here } I_D = 1A, C = 2 \times 10^{-6} F$$

27. **Ans (2)**

Parallel wires having current in same direction attract each other

$$F_{BC} > F_{BA}$$

28. **Ans (1)**

The current in the circuit

$$I = \frac{(20 - 10)}{(7.5 + 0.5 + 1 + 1)} = 1 \text{ Amp}$$

The potential difference across PS,

$$V_{PS} = 7.5 \times 1 = 7.5 \text{ V}$$

The potential difference across QR,

$$V_{QR} = 1 \times 1 = 1 \text{ V}$$

As point G is connected to earth hence potentials of R and S is zero.

The direction of the current in the circuit is from P to S, hence point P is at higher potential.

$$\therefore V_P = 7.5 \text{ V}$$

$$\text{Similarly } V_Q = -1 \text{ V}$$

29. Ans (2)

$$a = \frac{F}{m}$$

$$S = Ut + \frac{1}{2}at^2; S = 0 + \frac{1}{2}\left(\frac{F}{m}\right)t^2$$

$$\vec{W} = \vec{F} \cdot \vec{S}; W = (F) \left(\frac{Ft^2}{2m}\right)$$

$$W = \frac{F^2 t^2}{2m}$$

30. Ans (3)

$$v_B \cos \theta = v_A$$

for 'B' to remain north of 'A'

$$\frac{v_B}{v_A} = \frac{1}{\cos \theta} = \sec \theta$$

31. Ans (1)

$$\text{formula } \int \cos x \, dx = \sin x + c$$

32. Ans (2)

$$\mu \propto \frac{1}{\lambda}$$

33. Ans (2)

$$y = 5 \sin \frac{\pi x}{3} \cos 40\pi t$$

Comparing with the standard equation of stationary wave

$$y = A \sin kx \cos \omega t$$

$$\text{we get; } k = \frac{\pi}{3} = \frac{2\pi}{\lambda}$$

$$\Rightarrow \therefore \lambda = 6 \text{ cm}$$

Hence, the separation between two adjacent

$$\text{nodes} = \frac{\lambda}{2} = 3 \text{ cm}$$

34. Ans (2)

$$\Delta KE = \Delta PE$$

$$\frac{1}{2}mv^2 = \frac{mgh}{1 + \frac{h}{R}} \quad g = \frac{Gm}{R^2}$$

$$\frac{1}{2} \frac{Gm}{R} = \frac{Gm}{R^2} \times \frac{h}{1 + \frac{h}{R}}$$

$$R + h = 2h \Rightarrow h = R$$

35. Ans (1)

$$\text{Power factor} = \cos \phi = \frac{V_R}{V_{\text{net}}}$$

$$= \frac{80}{\sqrt{(80)^2 + (100 - 40)^2}}$$

$$= 0.8$$

36. Ans (4)

Distance between the first dark fringes on either side of central maxima = width of central maxima

$$= \frac{2\lambda D}{d} = \frac{2 \times 600 \times 10^{-9} \times 2}{1 \times 10^{-3}} = 2.4 \text{ mm.}$$

37. Ans (3)

Given $T_1 = 800 \text{ K}$ and $T_2 = 400 \text{ K}$,

$$W = Q_1 - Q_2 = 750 \text{ W}$$

Amount of heat absorbed is given by

$$\frac{W}{Q_1} = \frac{T_1 - T_2}{T_1} = \frac{800 - 400}{800} = \frac{1}{2}$$

$$\Rightarrow \frac{750}{Q_1} = \frac{1}{2}$$

$$\Rightarrow Q_1 = 1500 \text{ Watt. or J/sec}$$

$$Q_1 = \frac{1500}{4.2} = 357.14 \text{ cal/sec}$$

Efficiency

$$\eta = \frac{T_1 - T_2}{T_1} = \frac{800 - 400}{800} = \frac{1}{2} \text{ or } 50\%$$

38. Ans (3)

$$MP = \frac{-f_0}{f_e} \left(1 + \frac{f_e}{D}\right)$$

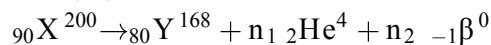
$$MP = \frac{-100}{10} \left(1 + \frac{10}{25}\right)$$

$$MP = -14$$

39. Ans (4)

There is no flux change through loop.

40. Ans (4)



Mass No. balance

$$200 = 168 + n_1 \cdot 4$$

$$\Rightarrow n_1 = 8$$

At. No. balance

$$90 = 80 + n_1 (2) + n_2 (-1)$$

$$\Rightarrow n_2 = 6$$

41. **Ans (3)**

Potential decreases in the direction of electric field, Hence
 $(V_A = V_B) > V_C$

42. **Ans (2)**

For every closed circular orbit of satellite

$$K.E = -\frac{P \cdot E}{2}$$

43. **Ans (1)**

for sliding cylinder

$$a_{\text{slide}} = g \sin \theta$$

for rolling cylinder

$$a_{\text{roll}} = \frac{g \sin \theta}{1 + \frac{1}{2}} = \frac{2}{3} g \sin \theta$$

Hence (1)

44. **Ans (1)**

$$v_{B/F} = \mu v_B + v_F = \frac{4}{3} \times 9 + 3$$

$$v_B = 15 \text{ m/sec.}$$

45. **Ans (3)**

$$\beta = \frac{\alpha}{1 - \alpha}$$

46. **Ans (4)**

$$\text{Heat of Hydrogenation} \propto \frac{1}{\text{stability of alkene}}$$

49. **Ans (2)**

$$m = \frac{\Delta_B \times 1000}{\Delta_A \times M_A}$$

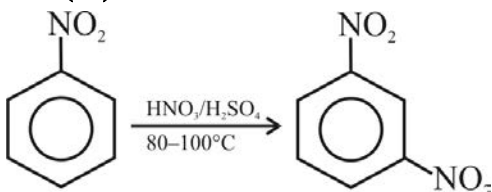
$$\frac{mM}{1000} A = \frac{\Delta_B}{n_A}$$

$$\frac{mM_A}{1000} = \frac{X_B}{1 - X_B}$$

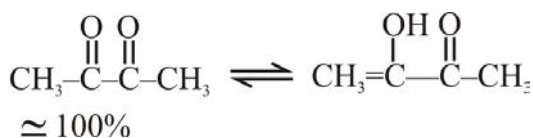
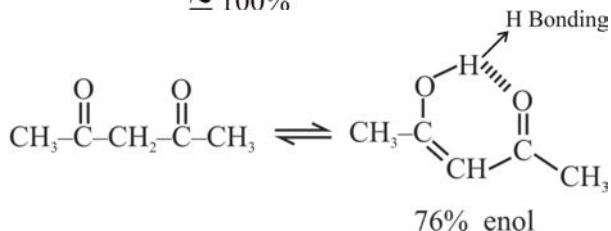
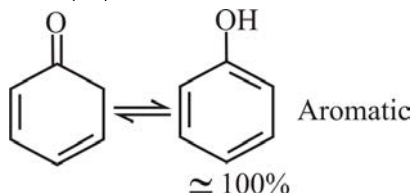
51. **Ans (4)**

$$d = \frac{ZM}{N_A \times a^3}$$

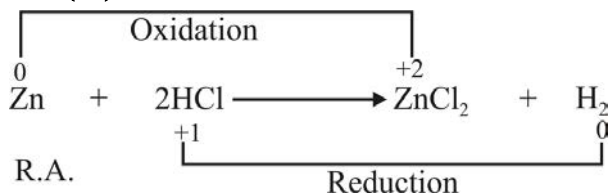
52. **Ans (3)**



54. **Ans (4)**



57. **Ans (3)**



58. **Ans (2)**

Fact

59. **Ans (2)**

$$\text{Number of electrons} = \frac{1.6}{16} \times 10 \times N_A = N_A$$

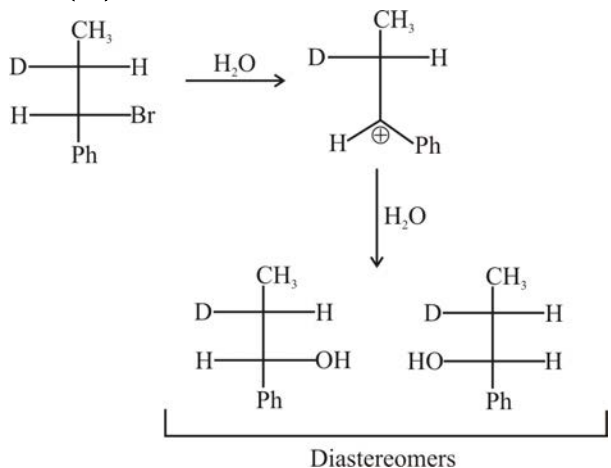
60. **Ans (3)**

'b' is same for a particular type of electrolyte.

61. **Ans (3)**

Fact

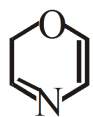
63. **Ans (2)**



64. Ans (3)

$$\Delta v = \frac{h}{4\pi m \Delta x}$$

66. Ans (3)



It has two heteroatoms

67. Ans (3)

Rate of nucleophilic addition reaction $\mu + ve$ charge density at carbonyl carbon.

70. Ans (3)

$$25 = \frac{1}{K} \times \frac{20}{4}$$

$$K = 1/5$$

$$\text{Now } \lambda_{eq} = \frac{1/5 \times 1000}{1/2} = 400 \Omega^{-1} \text{ cm}^2 \text{eq}^{-1}$$

72. Ans (4)

Fact

74. Ans (2)

Defination

75. Ans (3)

Insulin must have one 'S' atom

now 0.32% insulin weighs 32

$$\text{molecular mass} = \frac{32}{.32 \times 10^{-2}} = 10000$$

77. Ans (1)

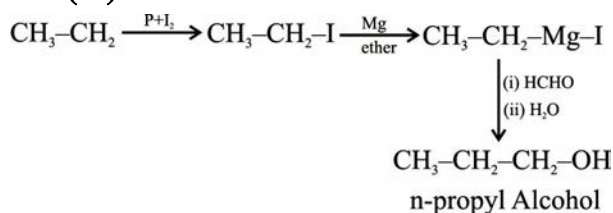
$$r = k[\text{OCI}^-][\text{I}^-][\text{OH}^-]^{-1}$$

order = -1 with respect to $[\text{OH}^-]$ ion

78. Ans (3)

$$\Delta S = nC_P \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1} = 2 \times \frac{5}{2} R \ln \frac{600}{300} + 2 \times R \ln \frac{2V}{V} = 7R \ln 2.$$

79. Ans (3)



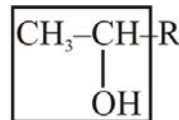
80. Ans (1)

$$K_p = K_c(RT)^2$$

$$T = 1/2R \Rightarrow T = 6.09 \text{ K}$$

81. Ans (4)

Conditions to give iodoform test



83. Ans (1)

When C = O bond is in conjugation than its rotational energy barrier decreases.

84. Ans (4)

Zinc gets oxidised most easily.

85. Ans (1)

NO = $3e^-$ donor

$\therefore$ 3 CO can be replaced by 2 NO

86. Ans (1)

$a \uparrow \Rightarrow$ intermolecular forces $\uparrow$

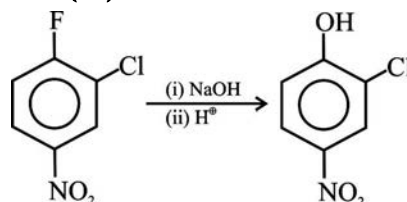
87. Ans (1)

NCERT Pg. # 101, $\text{O}_3 \Rightarrow$ structure

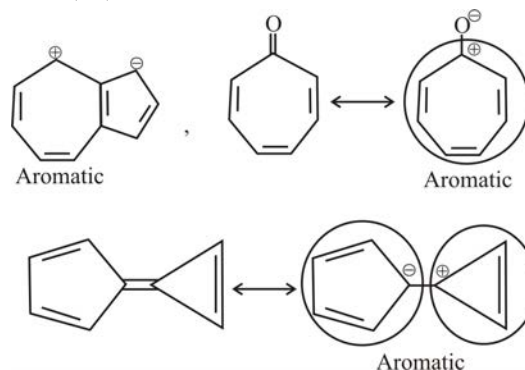
88. Ans (3)

$$\text{pK}_b = 14 - 9.25 = 4.75$$

89. Ans (2)



90. Ans (4)



- | | |
|---|--|
| 98. Ans (1)
NCERT Pg. # 60 | 147. Ans (2)
NCERT XII Pg. # 284 |
| 103. Ans (2)
NCERT XI Pg. # 106 | 151. Ans (2)
Nodal tissue is a mass of neuromuscular tissue.
eg. Sino atrial node (SA node) is pacemaker.
OR
NCERT XIth Pg.#284,287 |
| 115. Ans (1)
NCERT XIth Pg. # 96 | 154. Ans (3)
R.A.A.S → Renin → Angiotensin → Aldosterone
system. |
| 117. Ans (4)
NCERT Pg. # 165, 10.2.2 | 159. Ans (4)
NCERT-XI, Pg # 132, Para-4 |
| 122. Ans (1)
NCERT (XII th) Pg. # 194 (Para 11.1) | 160. Ans (1)
NCERT (XII) Pg. # 27 |
| 127. Ans (1)
NCERT (XII) Pg. # 37 | 163. Ans (2)
NCERT XI Pg # 146 (E), Table 9.3, 146(H) |
| 132. Ans (2)
NCERT Page-156 | 165. Ans (4)
NCERT (XII th) Pg. # 63 |
| 133. Ans (3)
NCERT Pg. # 7 | 166. Ans (4)
NCERT XI (E/H) Pg # 17, II Para |
| 136. Ans (1)
NCERT (XI th) Pg. # 263,264 | 174. Ans (2)
NCERT Pg.(XII) Page No. 49(E), 53(H) |
| 138. Ans (1)
NCERT (XI) Pg. # 80 | 178. Ans (2)
NCERT Page-09 |
| 144. Ans (1)
NCERT XI th Pg.#298 Para-4 | |